

HE2003 Econometrics I

Tutorial 4

09/09/2024

Question 1

Consider the following example of studying the relationship between test scores and class sizes in primary schools:

$$TestScore = \beta_0 + \beta_1 STR + u$$

We consider three different scenarios:

- (1) The kids are randomly enrolled to nearby primary schools.
- (2) The kids are selectively enrolled: the kids with high ability will be enrolled to good schools and those with low ability will be enrolled to bad schools.
- (3) The kids' parents decide which schools to apply. In the case that the number of applicants exceeds the vacancy of the school, a random balloting will be implemented.

Discuss whether the OLS Assumption 1 is likely to hold for each scenario.

Question 1

***These are open questions, so it is fine as long as the arguments are reasonable.**

(1) The kids are randomly enrolled to nearby primary schools.

The random assignment of kids implies that STR and $ability$ will be independent from each other. In this case, $E[ability|STR] = E[ability]$. Furthermore, we have $E[ability] = 0$ as we can always assume $E[u] = 0$, where $E[u]$ denotes the unconditional mean of the error term u . Therefore, OLS Assumption 1 will hold in this case.

(2) The kids are selectively enrolled: the kids with high ability will be enrolled to good schools and those with low ability will be enrolled to bad schools.

If good schools tend to have low STR (as they have more resources to hire more teachers), while bad schools tend to have high STR , the selection with regard to kids' abilities implies that $ability$ and STR will be negatively correlated (on average $ability$ is low when STR is high, and vice versa). Therefore, OLS Assumption 1 will be violated in this case.

Question 1

(3) The kids' parents decide which schools to apply. In the case that the number of applicants exceeds the vacancy of the school, a random balloting will be implemented.

It is reasonable to argue that all parents will first try the good schools (assuming they all care about their kids). In this case, a random balloting will occur as the number of applicants will exceed the vacancy of the good schools. As a consequence of the random balloting, kids' with different levels of ability will have the same chances to be enrolled into good schools and bad schools, so that OLS Assumption 1 is likely to hold.

Question 2

- (a) Explain the measure of fit: R^2 .
- (b) Illustrate the meaning of explained sum of squares (ESS) and total sum of squares (TSS) using a diagram.

R^2 measures the fraction of the sample variation in Y that is explained by X . Notice that Y_i can be decomposed as:

$$Y_i = \hat{Y}_i + \hat{u}_i$$

where $\hat{Y}_i$ is the OLS predicted value and $\hat{u}_i$ is the residual. R^2 is the ratio of the sample variance of $\hat{Y}$ to the sample variance of Y .

Specifically, the R^2 can be written as the ratio of the explained sum of squares (ESS) to the total sum of squares (TSS):

$$R^2 = \frac{ESS}{TSS} , \quad ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 , \quad TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

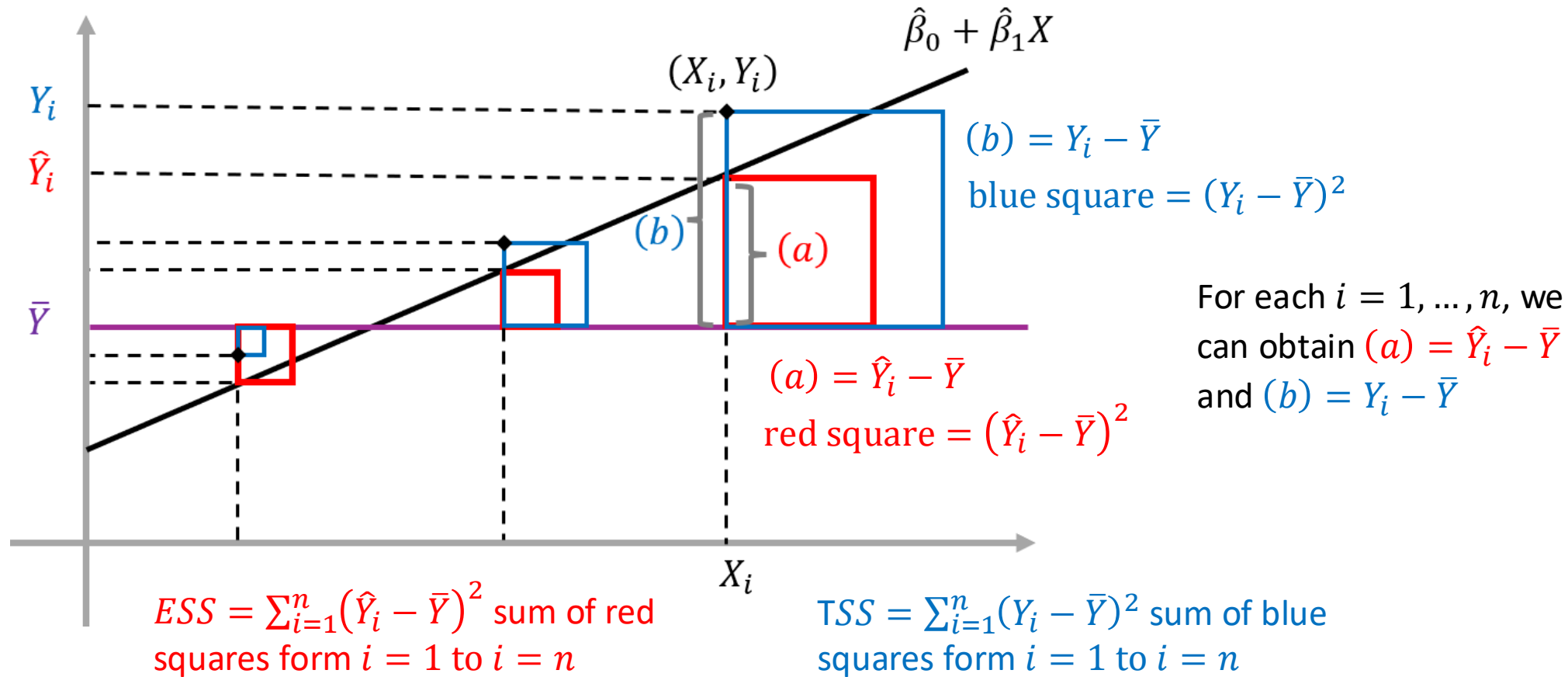
Question 2

ESS is equal to the sum over $i = 1, \dots, n$ of the square of the difference between $\hat{Y}_i$ and $\bar{Y}$ ($\hat{Y}_i - \bar{Y}$, part (a) in the following graph)

TSS is equal to the sum over $i = 1, \dots, n$ of the square of the difference between Y_i and $\bar{Y}$ ($Y_i - \bar{Y}$, part (b) in the following graph).

The difference between Y_i and $\hat{Y}_i$ is the residual $\hat{u}_i$. Furthermore, although for some i , $(\hat{Y}_i - \bar{Y})^2$ can be larger than $(Y_i - \bar{Y})^2$, we always have $ESS \leq TSS$ overall.

Question 2



Question 3

- (a) Show that the sum of the OLS residuals $\hat{u}_i$ ($\sum_{i=1}^n \hat{u}_i$) is equal to zero by using the first-order condition (FOC) of the OLS estimators.
- (b) Show that the average of the OLS predicted values $\hat{Y}_i$ ($\frac{1}{n} \sum_{i=1}^n \hat{Y}_i$) is equal to the sample average of Y ($\bar{Y}$).
- (c) Show that if the OLS estimator $\hat{\beta}_1 = 0$, then $R^2 = 0$.

Question 3

(a) Show that the sum of the OLS residuals $\hat{u}_i$ ($\sum_{i=1}^n \hat{u}_i$) is equal to zero by using the first-order condition (FOC) of the OLS estimators.

Recall the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the solution of the minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$$

We have the first equation of the FOC (taking partial derivatives with respect to b_0):

$$\sum_{i=1}^n [Y_i - (\underbrace{\hat{\beta}_0 + \hat{\beta}_1 X_i}_{\hat{Y}_i})] = 0$$

Hence, $\sum_{i=1}^n \hat{u}_i = 0$

Question 3

(b) Show that the average of the OLS predicted values $\hat{Y}_i$ ($\frac{1}{n} \sum_{i=1}^n \hat{Y}_i$) is equal to the sample average of Y ($\bar{Y}$).

Solution 1:

The actual value of Y_i can be decomposed into the OLS predicted value $\hat{Y}_i$ and the residual $\hat{u}_i$.

$$Y_i = \hat{Y}_i + \hat{u}_i$$

$$\begin{aligned}\bar{Y} &= \frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i + \hat{u}_i) \\ &= \frac{1}{n} \sum_{i=1}^n \hat{Y}_i + \frac{1}{n} \sum_{i=1}^n \hat{u}_i = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i \\ &\quad = 0 \text{ from part (a)}\end{aligned}$$

Solution 2:

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n \hat{Y}_i &= \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i) \\ &= \frac{1}{n} \sum_{i=1}^n \hat{\beta}_0 + \frac{1}{n} \sum_{i=1}^n \hat{\beta}_1 X_i = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}\end{aligned}$$

According to OLS formula: $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y} - \hat{\beta}_1 \bar{X} + \hat{\beta}_1 \bar{X} = \bar{Y}$$

Question 3

Show that if the OLS estimator $\hat{\beta}_1 = 0$, then $R^2 = 0$.

Solution 1:

If $\hat{\beta}_1 = 0$ then we have $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i = \hat{\beta}_0$

In addition, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = \bar{Y}$ (OLS formula)

Combining these two equations $\Rightarrow \hat{Y}_i = \bar{Y}$

$$\Rightarrow ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \sum_{i=1}^n 0 = 0$$

$$\Rightarrow R^2 = \frac{ESS}{TSS} = 0$$

Solution 2:

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

$$ESS = \sum_{i=1}^n ((\hat{\beta}_0 + \hat{\beta}_1 X_i) - (\hat{\beta}_0 + \hat{\beta}_1 \bar{X}))^2$$

$$ESS = \sum_{i=1}^n (\hat{\beta}_1 X_i - \hat{\beta}_1 \bar{X})^2 = \sum_{i=1}^n (\hat{\beta}_1 (X_i - \bar{X}))^2$$

If $\hat{\beta}_1 = 0$, then $ESS = 0$

Question 4

A random sample of 100 20-year-old men is selected from a population and these men's height and weight are recorded. An OLS regression of weight on height yields:

$$\widehat{Weight} = -79.24 + 4.16 \times Height, \quad R^2 = 0.72$$

where *Weight* is measured in pounds and *Height* is measured in inches.

- (a) What is the regression's weight prediction for someone who is 64 inches tall? 68 inches tall? 72 inches tall?
- (b) A man has a late growth spurt and grows 2 inches over the course of a year. What is the regression's prediction for the increase in this man's weight?
- (c) Suppose that instead of measuring weight and height in pounds and inches, these variables are measured in centimeters and kilograms. What are the OLS estimates from this new centimeter-kilogram regression? How about the R^2 ?

Question 4

(a) What is the regression's weight prediction for someone who is 64 inches tall? 68 inches tall? 72 inches tall?

For 64 inches tall, $\widehat{Weight} = -79.24 + 4.16 \times 64 = 187$ pounds.

For 68 inches tall, $\widehat{Weight} = -79.24 + 4.16 \times 68 = 203.64$ pounds.

For 72 inches tall, $\widehat{Weight} = -79.24 + 4.16 \times 72 = 220.28$ pounds.

(b) A man has a late growth spurt and grows 2 inches over the course of a year. What is the regression's prediction for the increase in this man's weight?

$$\Delta \widehat{Weight} = 4.16 \times \Delta Height = 4.16 \times 2 = 8.32 \text{ (pounds)}$$

Question 4

(c) Suppose that instead of measuring weight and height in pounds and inches, these variables are measured in centimeters and kilograms. What are the OLS estimates from this new centimeter-kilogram regression? How about the R^2 ?

According to the formula in the last tutorial, if we multiply X by c_1 and Y by c_2 , where c_1 and c_2 are constants, then we have

$$\tilde{\beta}_1 = \left(\frac{c_2}{c_1} \right) \hat{\beta}_1 \quad \text{and} \quad \tilde{\beta}_0 = c_2 \hat{\beta}_0$$

where $(\hat{\beta}_0, \hat{\beta}_1)$ denote the original OLS estimators computed from regression of Y on X , while $(\tilde{\beta}_0, \tilde{\beta}_1)$ denote the new OLS estimators computed from regression of $c_2 Y$ on $c_1 X$.

Question 4

We have 1 inch = 2.54 cm, and 1 lb = 0.4536 kg. Hence, in this case $c_1 = 2.54$ and $c_2 = 0.4536$.

Therefore, in the new centimeter-kilogram regression, the new slope estimate should be:

$$\left(\frac{0.4536}{2.54}\right) \times 4.16 = 0.7429$$

and the new intercept estimate should be:

$$0.4536 \times (-79.24) = -35.9433$$

Furthermore, we can verify that if we multiply X by c_1 and Y by c_2 , the R^2 remains unchanged. The R^2 is in fact unit free. Thus, R^2 is still equal to 0.72.

Question 4

We have:

$$\tilde{Y}_i = \tilde{\beta}_0 + \tilde{\beta}_1 \tilde{X}_i = c_2 \hat{\beta}_0 + \left(\frac{c_2}{c_1} \right) \hat{\beta}_1 \times c_1 X_i = c_2 (\hat{\beta}_0 + \hat{\beta}_1 X_i) = c_2 \hat{Y}_i$$

Let $\tilde{R}^2$ be the new R^2 computed from regression of $c_2 Y$ on $c_1 X$. Then:

$$\tilde{R}^2 = \frac{\sum_{i=1}^n (\tilde{Y}_i - \tilde{\bar{Y}})^2}{\sum_{i=1}^n (\tilde{Y}_i - \tilde{\bar{Y}})^2} = \frac{\sum_{i=1}^n (c_2 \hat{Y}_i - c_2 \bar{Y})^2}{\sum_{i=1}^n (c_2 Y_i - c_2 \bar{Y})^2} = \frac{c_2^2 \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{c_2^2 \sum_{i=1}^n (Y_i - \bar{Y})^2} = R^2 \text{ (original)}$$